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IEEE INFOTHEORY INDUSTRY CONCLAVE 2026 | THEORY ➔ TECHNOLOGY ➔ IMPACT

CHRIST (Deemed to be University) Yeshwanthpur Campus, Bengaluru

VVIT (VTU) Bengaluru - 77

ROUTE RIOT AI-Based Smart Bus Occupancy Prediction and Passenger Redirection System

Bridging Hybrid Edge Sensing (YOLO & ToF) with Predictive AI & Multi-Modal Redirection for Bengaluru Transport

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01 ABSTRACT & INTRODUCTION

Bengaluru's public transport network serves an immense, constantly shifting commuter demand. While existing transit apps show vehicle GPS coordinates and ETAs, commuters remain **completely unaware of in-cabin crowd density** and impending congestion spikes before boarding.

Route Riot is an intelligent public-transport assistance framework that combines **YOLO-based computer vision** for camera-equipped buses with **ESP32 + Time-of-Flight (ToF) sensors** for non-camera fleets. Occupancy streams are normalized through a high-throughput **FastAPI backend** to establish live passenger states and forecast upcoming surges.

SENSE ➔ DETECT ➔ PREDICT ➔ REDIRECT

Camera / ToF ➔ Common Event ➔ AI Forecast ➔ Multi-Modal Alternative

KEY NOVELTIES:

- **Dual Sensing:** Bridges advanced AI cameras with low-cost ToF hardware.
- **Predictive Layer:** Anticipates stop-by-stop crowding 15–20 minutes prior.
- **Multi-Modal Sync:** Directs passengers to parallel BMTC routes and Namma Metro.

02 PROBLEM STATEMENT & SCENARIOS

THE CORE PROBLEM (SLIDE 5):
Bengaluru's bus passengers often do not know the actual crowd level of an approaching bus or how crowded it may become during their journey. *A bus being nearby does not necessarily mean it is the best travel choice.*

⚠️ **Normal Scenario:**

- Passenger wants to travel
- Bus location is available & ETA is known
- Passenger boards arriving bus blindly
- Actual in-cabin crowd level is unknown
- **Result: Severe overcrowding & discomfort**

✅ **Crowd-Uncertainty / Route Riot:**

- Bus is already highly occupied (>70%)
- More passengers expected at next stops
- Nearby alternate bus (500K) has lower crowd
- Namma Metro option is also available
- **Result: Informed, comfortable journey**

Blind Commute: Unpredictable boarding, severe fatigue, missed alternatives.

Route Riot: Pre-boarding clarity, optimal comfort, multi-modal flexibility.

Takeaway: Without predictive occupancy information, passengers miss better available alternatives departing just minutes apart.

03 KEY PROJECT OBJECTIVES

1. Real-Time Awareness
Provide clear, instant bus crowd conditions to eliminate daily commuter uncertainty.

2. Predictive Management
Analyze historical patterns to anticipate crowd spikes before they become severe.

3. Intelligent Redirection
Help passengers respond to crowded buses by identifying suitable BMTC & Metro routes.

4. Hybrid Fleet Framework
Unify camera-equipped (YOLO) and non-camera (ToF) vehicles into one common metric.

5. Improved Travel Experience
Reduce inconvenience of overcrowded cabins through actionable passenger decision support.

6. Transport Intelligence
Build a robust dataset supporting future AI transit demand analysis and smart fleet planning.

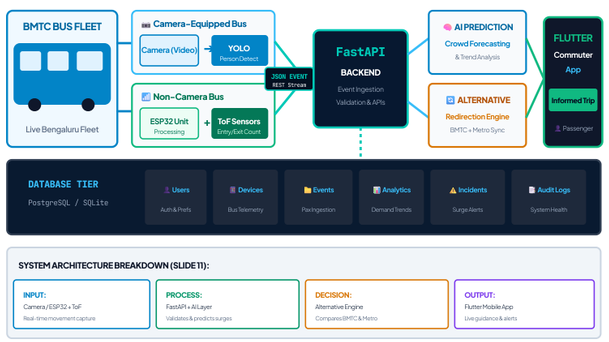
⚡ **Latency Target:** <1.5s Sensor-to-App 🎯 **Prediction Precision:** >88% 💰 **Fleet Retrofit:** <₹3,500/Bus

04 LITERATURE SURVEY & GAPS

Study / Domain	Parameters / Focus	Identified Research Gap
Vision Counting (2022)	Camera-based passenger counting	Occlusion in dense crowds reduces detection accuracy.
YOLO People Detection (2023)	Object detection for passenger count	Detection alone does not predict future route crowding.
IoT Distance Sensing (2021)	ToF doorway entry/exit counting	Cannot resolve multi-door simultaneous passenger boarding.
ML Demand Forecast (2023)	Historical ridership & passenger flow	Demand model is separated from real-time vehicle occupancy.
Smart Mobility Apps (2024)	Real-time transit info & ETA tracking	Lacks integration between crowd prediction and multi-modal transit.

💡 **Research Contribution:** Route Riot closes these gaps by unifying hybrid edge sensing, common event normalization, proactive AI surge prediction, and live multi-modal redirection.

05 ROUTE RIOT SYSTEM ARCHITECTURE



Hybrid Ingestion: Camera fleets process passenger density via edge YOLO inference; legacy fleets utilize dual doorway ToF break-beam sensors. All streams feed into a unified schema for real-time validation.

06 METHODOLOGY & 6-STAGE PIPELINE

1. SENSING Camera / YOLO & ESP32 + ToF

2. PROCESSING FastAPI Validation & Storage

3. PREDICTION Low / Mod / High Surge AI

4. REDIRECTION BMTC, Metro & Feeder sync

1. CAPTURE Camera & ToF

2. PROCESS YOLO count

3. INGEST FastAPI JSON

4. PREDICT Crowd models

5. RECOMMEND Alternative routes

6. INFORM Flutter app UI

```
EVENT: {"bus_id": "KA01-500D", "route": "ORR", "curr_pax": 42, "cap": 60,
"load_pct": 70, "method": "YOLO"}                                  [VALIDATED]
```

07 HARDWARE & SOFTWARE SPECIFICATIONS

HARDWARE STACK (SLIDE 8):

ESP32:	Processes ToF data & communicates
ToF Sensors:	VL53L0X doorway entry/exit detection
Camera:	Captures in-cabin passenger video
Edge Node:	Runs YOLO inference on camera buses
Smartphone:	Runs Route Riot Flutter mobile app

SOFTWARE STACK (SLIDE 8):

Flutter/Dart:	Mobile application development
Python/FastAPI:	REST API & backend communication
YOLO:	Person detection & occupancy
AI/ML Models:	Crowd classification & surge forecast
Firebase/SQLite:	Cloud DB & historical data storage

Communication: REST / JSON & WebSockets **Cost Profile:** Ultra-Low (<₹3,500/non-camera bus retrofit)

08 3-TIER CROWD CLASSIFICATION & ROUTE CURVE



Real-Time Bus Occupancy Estimation
YOLO & ToF provide unified load estimate across all configurations:
Standard 12m | Articulated | Mini Feeder

🟢 **LOW** 0 – 30% Plenty of space. Travel comfortably.

🟡 **MODERATE** 31 – 70% Getting busier. Plan ahead if possible.

🔴 **HIGH** 71 – 100% Very crowded. Consider alternative.

PREDICTED CROWD LEVEL ALONG THE ROUTE (SLIDE 12): Next Crowding Peak: 08:45 AM (Yeshwanthpur)

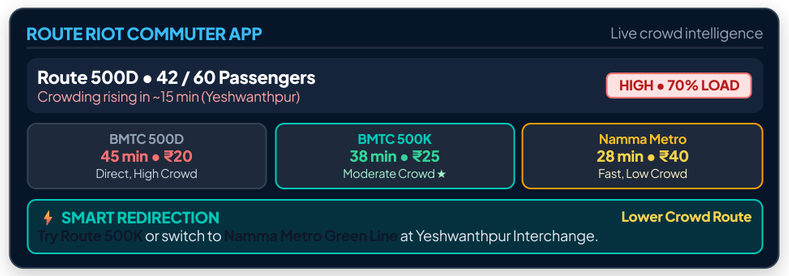


Route View: Majestic (Low) ➔ Shivajinagar (Mod) ➔ Sadashivanagar (Mod) ➔ Yeshwanthpur **(High)** ➔ Hebbal (Mod) ➔ RT Nagar (Low)

The Result: Earlier awareness, better choices, avoid peak crowd.

09 MULTI-MODAL REDIRECTION & COMMUTER APP

When an approaching bus is crowded, the **Alternative Engine** evaluates multi-modal travel options to provide instant, stress-free travel recommendations:



Passenger Decision Support (Slide 13): Deliver actionable information through Flutter. Passengers view Live Occupancy + Crowd Level + Prediction + Multi-Modal Alternatives in one interface.

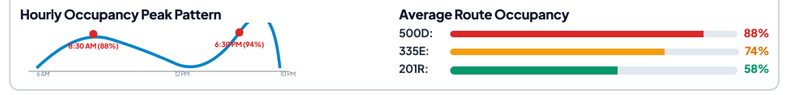
10 TRANSPORT ANALYTICS FOR BMTC

1. Peak-Demand Identification
Pinpoint exact stops and times where crowding builds up across corridors.

2. Route-Level Crowd Analysis
Compare average occupancy across parallel routes for transit load-balancing.

3. Future Model Retraining
Continuous learning using real-world sensor logs to boost forecast accuracy.

4. Dynamic Fleet Allocation
Data-driven dispatch for extra feeder buses to alleviate choke points.



11 CONCLUSION & MAJOR REFERENCES

🎯 **Conclusion (Slide 15):** Route Riot combines IoT sensors & AI to monitor and predict bus occupancy in real time. Through the Flutter application, passengers can view crowd levels and make informed travel decisions. The system aims to reduce overcrowding, improve passenger distribution, and make public transportation smarter and more efficient.

MAJOR REFERENCES (SLIDE 14):

1. **Hsu et al.** — Estimation of the Number of Passengers in a Bus Using Deep Learning.
2. **Wood et al.** — Frameworks for Real-Time Bus Passenger Occupancy Prediction.
3. **Dharmarajan et al.** — Depth Imaging-Based Passenger Counting Sensor for Public Transit.
4. **Li et al.** — Automatic Passenger Counting System for Bus Based on RGB-D Video.
5. **Transp. Res. Procedia** — Wireless Sensor Network Architecture for Passenger Counting.
6. **Ultralytics** — YOLO Documentation; **STMicroelectronics** — VL53L0X ToF Sensor Documentation.

CONCLAVE IMPACT **Route Riot: Moving beyond bus tracking toward predictive, multi-modal passenger assistance.**

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IEEE Information Theory Industry Conclave • October 3, 2026

PROJECT REPO
Scan for project code, FastAPI & Flutter demo